

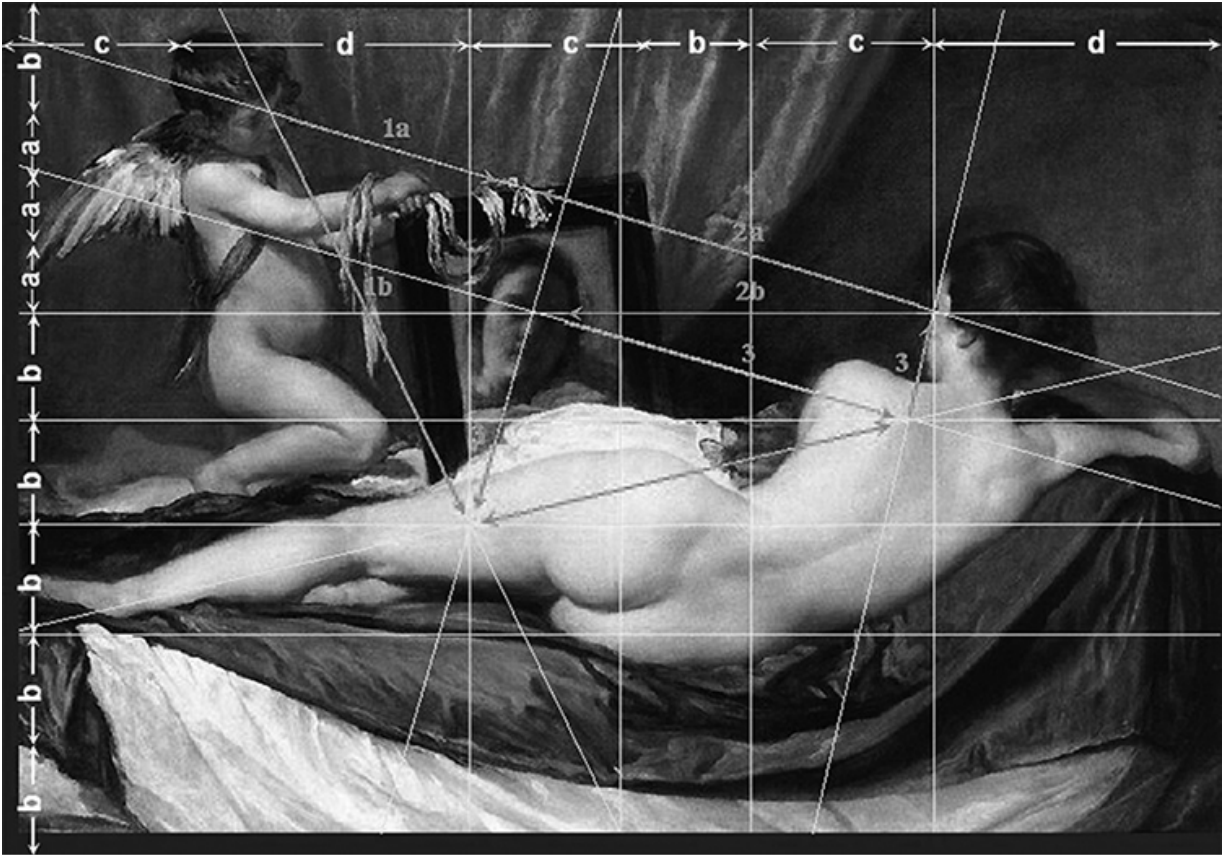


Figure 5-1: Phi/Venus (from https://commons.wikimedia.org/wiki/File:DV_The_Toilet_of_Venus_Gr.jpg)

Compressing and Communicating Phi

Phi's exact value is surprisingly hard to express. I could say that it's equal to $1.61803399 \dots$. The ellipsis here is a way of cheating; it means that more numbers follow (an infinite number of numbers, in fact), but I haven't told you what those numbers are, so you still don't know the exact value of phi.

For some numbers with infinite decimal expansions, a fraction can represent them exactly. For example, the number $0.11111 \dots$ is equal to $1/9$ —here, the fraction provides an easy way to express the exact value of an infinitely continued decimal. Even if you didn't know the fractional representation, you could see the pattern of repeating 1s in $0.1111 \dots$ and thereby understand its exact value. Unfortunately, the golden ratio is what's called an *irrational number*, meaning that there are no two integers x and y